Full Stack Assignment – 16

Assignment ID: FSD-16

Title: Document Access Control System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Document Access Control System** that allows users to request access to documents and administrators to manage document approvals.
The system must enforce approval-based access and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- View available documents
- Request access to documents
- Download approved documents

ADMIN

- Log in to the system
- Upload and manage documents
- Approve or reject document access requests

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Document Management

- ADMIN can upload documents with metadata
- Document details must be stored in the database
- Document access status must be maintained persistently

Access Control

- USER can request access to a document
- ADMIN can approve or reject access requests

- Only approved documents should be downloadable by users

4️⃣ Business Rule (Critical)

⚠️ **Only documents with approved access may be downloaded by users.**

Any download attempt without approval must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All document and access data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Sensitive configuration must use environment variables

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Document listing with access status
- Access request and download actions

UI styling is not evaluated; access control logic matters most.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions

- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **authorization correctness, rule enforcement, and end-to-end execution**.